

EXPERIMENT NO : 07

Random Sampling and Sampling Distribution

PROGRAM:-

```
import numpy as np
import matplotlib.pyplot as plt
population_mean = 50
population_std = 10
population_size = 100000

population = np.random.normal(
    population_mean,
    population_std,
    population_size
)

sample_sizes = [30, 50, 100]
num_samples = 1000

sample_means = {}

for size in sample_sizes:
    sample_means[size] = []

    for _ in range(num_samples):
        sample = np.random.choice(
            population,
            size=size,
```

```
        replace=False
    )
    sample_means[size].append(np.mean(sample))
plt.figure(figsize=(12, 8))

for i, size in enumerate(sample_sizes):
    plt.subplot(len(sample_sizes), 1, i + 1)

    plt.hist(
        sample_means[size],
        bins=30,
        alpha=0.7,
        label=f"Sample Size {size}"
    )

    plt.axvline(
        np.mean(population),
        color="red",
        linestyle="dashed",
        linewidth=1.5,
        label="Population Mean"
    )

plt.title(f"Sampling Distribution (Sample Size {size})")
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
```

```
plt.legend()
```

```
plt.tight_layout()
```

```
plt.show()
```

OUTPUT:-

